

# Signals & System

## Lecture 14

### Laplace Transform Properties

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# Lecture 14: Laplace Transform Properties

**Laplace transform** (3 lectures):

Laplace transform as Fourier transform with convergence factor. **Properties of the Laplace transform**

Specific objectives for today:

- Linearity and time shift properties
- Convolution property
- Time domain differentiation & integration property
- Transforms table

# Lecture 14: Resources

## Core material

SaS, O&W, Chapter 9.5&9.6

## Recommended material

MIT, Lecture 18

Laplace transform properties are very similar to the properties of a Fourier transform, when  $s=j\omega$

# Reminder: Laplace Transforms

Equivalent to the Fourier transform when  $s=j\omega$

$$X(s) = \int_{-\infty}^{\infty} x(t)e^{-st} dt \quad \text{Laplace transform}$$

$$x(t) = \frac{1}{2\pi j} \int_{\sigma-j\infty}^{\sigma+j\infty} X(s)e^{st} ds \quad \text{Inverse Laplace transform}$$

$$x(t) \overset{L}{\longleftrightarrow} X(s)$$

Associated region of convergence for which the integral is finite

Used to understand the frequency characteristics of a signal (system)

Used to solve ODEs because of their convenient calculus and convolution properties (today)

# Linearity of the Laplace Transform

If  $x_1(t) \xleftrightarrow{L} X_1(s) \quad \text{ROC} = R_1$

and  $x_2(t) \xleftrightarrow{L} X_2(s) \quad \text{ROC} = R_2$

Then  $ax_1(t) + bx_2(t) \xleftrightarrow{L} aX_1(s) + bX_2(s) \quad \text{ROC} = R_1 \cap R_2$

This follows directly from the definition of the Laplace transform (as the integral operator is linear). It is easily extended to a linear combination of an arbitrary number of signals

# Time Shifting & Laplace Transforms

If  $x(t) \xleftrightarrow{L} X(s)$  ROC=R

Then  $x(t-t_0) \xleftrightarrow{L} e^{-st_0} X(s)$  ROC=R

Proof  $x(t) = \frac{1}{2\pi j} \int_{\sigma-j\infty}^{\sigma+j\infty} X(s) e^{st} ds$

Now replacing  $t$  by  $t-t_0$

$$x(t-t_0) = \frac{1}{2\pi j} \int_{\sigma-j\infty}^{\sigma+j\infty} X(s) e^{s(t-t_0)} ds$$

$$= \frac{1}{2\pi j} \int_{\sigma-j\infty}^{\sigma+j\infty} (e^{-st_0} X(s)) e^{st} ds$$

Recognising this as

$$L\{x(t-t_0)\} = e^{-st_0} X(s)$$

A signal which is shifted in time may have both the **magnitude** and the **phase** of the Laplace transform altered.

# Example: Linear and Time Shift

Consider the signal (linear sum of two time shifted sinusoids)

$$x(t) = 2x_1(t - 2.5) - 0.5x_1(t - 4)$$

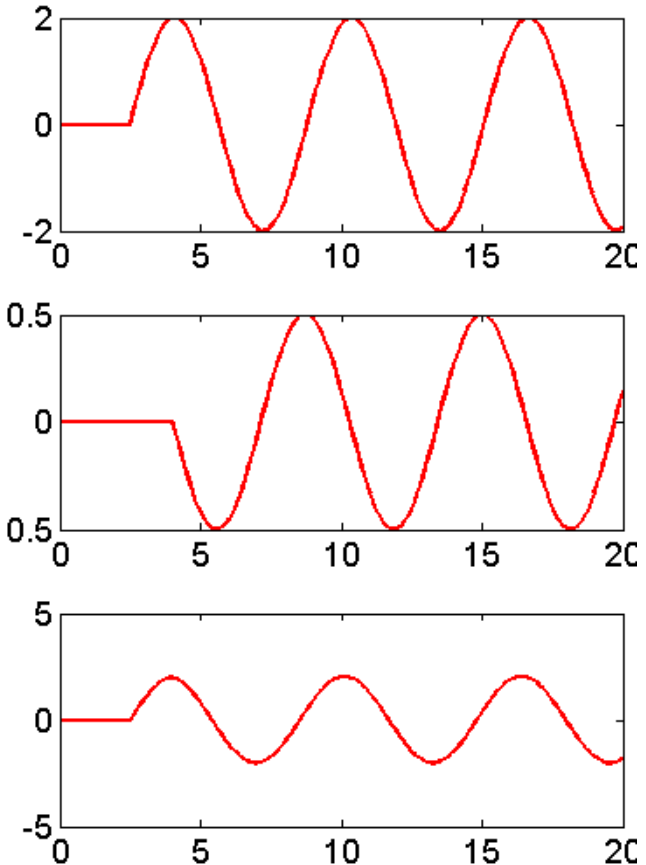
where  $x_1(t) = \sin(\omega_0 t)u(t)$ .

Using the sin() Laplace transform example

$$X_1(s) = \frac{\omega_0}{s^2 + \omega_0^2} \quad \text{Re}\{s\} > 0$$

Then using the **linearity** and **time shift** Laplace transform properties

$$X(s) = \left(2e^{-2.5s} - 0.5e^{-4s}\right) \frac{\omega_0}{s^2 + \omega_0^2} \quad \text{Re}\{s\} > 0$$



# Convolution

The Laplace transform also has the multiplication property, i.e.

$$\begin{array}{ll} \overset{L}{x(t)} \leftrightarrow X(s) & \text{ROC} = R_1 \\ \underset{L}{h(t)} \leftrightarrow H(s) & \text{ROC} = R_2 \\ x(t) * h(t) \leftrightarrow X(s)H(s) & \text{ROC} \supseteq R_1 \cap R_2 \end{array}$$

Proof is “identical” to the Fourier transform convolution

Note that pole-zero cancellation may occur between  $H(s)$  and  $X(s)$  which extends the ROC

$$X(s) = \frac{s+1}{s+2} \quad \Re\{s\} > -2$$

$$H(s) = \frac{s+2}{s+1} \quad \Re\{s\} > -1$$

$$X(s)H(s) = 1 \quad -\infty < \Re\{s\} < \infty$$



# Example 1: 1<sup>st</sup> Order Input & First Order System Impulse Response

Consider the Laplace transform of the output of a first order system when the input is an exponential (decay?)

$$x(t) = e^{-at}u(t)$$

$$h(t) = e^{-bt}u(t)$$

*Solved with Fourier transforms when  $a, b > 0$*

Taking Laplace transforms

$$X(s) = \frac{1}{s+a} \quad \text{Re}\{s\} > -a$$

$$H(s) = \frac{1}{s+b}, \quad \text{Re}\{s\} > -b$$

Laplace transform of the output is

$$Y(s) = \frac{1}{s+a} \frac{1}{s+b} \quad \text{Re}\{s\} > \max\{-a, -b\}$$

# Example 1: Continued ...

Splitting into partial fractions

$$Y(s) = \left( \frac{1}{b-a} \right) \left( \frac{1}{s+a} - \frac{1}{s+b} \right) \quad \text{Re}\{s\} > \max\{-a, -b\}$$

and using the inverse Laplace transform

$$y(t) = \frac{1}{b-a} \left( e^{-at} u(t) - e^{-bt} u(t) \right)$$

Note that this is the same as was obtained earlier, expect it is **valid for all**  $a$  &  $b$ , i.e. we can use the Laplace transforms to solve ODEs of LTI systems, using the system's impulse response

$$h(t) \overset{L}{\longleftrightarrow} H(s)$$

## Example 2: Sinusoidal Input

Consider the 1<sup>st</sup> order (possible unstable) system response with input  $x(t)$

$$h(t) = e^{-at}u(t)$$

$$x(t) = \cos(\omega_0 t)u(t)$$

Taking Laplace transforms

$$H(s) = \frac{1}{s+a} \quad \text{Re}\{s\} > -a$$

$$X(s) = \frac{s}{s^2 + \omega_0^2} \quad \text{Re}\{s\} > 0$$

The Laplace transform of the output of the system is therefore

$$\begin{aligned} Y(s) &= \frac{s}{s^2 + \omega_0^2} \frac{1}{s+a} \quad \text{Re}\{s\} > \max\{0, -a\} \\ &= \left( \frac{1}{a^2 + \omega_0^2} \right) \frac{as + \omega_0^2}{s^2 + \omega_0^2} + \left( \frac{-a}{a^2 + \omega_0^2} \right) \frac{1}{s+a} \end{aligned}$$

and the inverse Laplace transform is

$$y(t) = \frac{u(t)}{a^2 + \omega_0^2} \left( a \sin(\omega_0 t) + \omega_0 \cos(\omega_0 t) - ae^{-at} \right)$$

# Differentiation in the Time Domain

Consider the Laplace transform derivative in the time domain

$$x(t) \overset{L}{\longleftrightarrow} X(s) \quad \text{ROC} = R$$

$$x(t) = \frac{1}{2\pi j} \int_{\sigma-j\infty}^{\sigma+j\infty} X(s) e^{st} ds$$

$$\frac{dx(t)}{dt} = \frac{1}{2\pi j} \int_{\sigma-j\infty}^{\sigma+j\infty} sX(s) e^{st} ds$$

$$\frac{dx(t)}{dt} \overset{L}{\longleftrightarrow} sX(s) \quad \text{ROC} \supseteq R$$

$sX(s)$  has an extra zero at 0, and may cancel out a corresponding pole of  $X(s)$ , so ROC may be larger

Widely used to solve when the system is described by LTI differential equations

# Example: System Impulse Response

Consider trying to find the system response (potentially unstable) for a second order system with an impulse input  $x(t)=\delta(t)$ ,  $y(t)=h(t)$

$$a \frac{d^2 y(t)}{dt^2} + b \frac{dy(t)}{dt} + cy(t) = x(t)$$

Taking Laplace transforms of both sides and using the linearity property

$$aL\left\{\frac{d^2 y(t)}{dt^2}\right\} + bL\left\{\frac{dy(t)}{dt}\right\} + cL\{y(t)\} = L\{\delta(t)\}$$

$$L\{y(t)\}(as^2 + bs + c) = 1$$

$$L\{y(t)\} = H(s) = \frac{1}{as^2 + bs + c} = \frac{1}{a(s - r_1)(s - r_2)} = \frac{k_1}{(s - r_1)} + \frac{k_2}{(s - r_2)}$$

where  $r_1$  and  $r_2$  are distinct roots, and calculating the inverse transform

$$y(t) = k_1 e^{r_1 t} u(t) + k_2 e^{r_2 t} u(t)$$

The general solution to a second order system can be expressed as the sum of two complex (possibly real) exponentials

# Lecture 14: Summary

Like the Fourier transform, the Laplace transform is **linear** and represents time shifts  $(t-T)$  by multiplying by  $e^{-sT}$

## Convolution

$$x(t) * h(t) \stackrel{L}{\leftrightarrow} X(s)H(s) \quad \text{ROC} \supseteq R_1 \cap R_2$$

Convolution in the time domain is equivalent to multiplying the Laplace transforms

Laplace transform of the system's impulse response is very important  $H(s) = \int h(t)e^{-st}dt$ . Known as the **transfer function**.

## Differentiation

$$\frac{dx(t)}{dt} \stackrel{L}{\leftrightarrow} sX(s) \quad \text{ROC} \supseteq R$$

Very important for solving ordinary differential equations

# Questions

## Theory

SaS, O&W, Q9.29-9.32

Work through slide 12 for the first order system

$$a \frac{dy(t)}{dt} + by(t) = \delta(t)$$

Where the aim is to calculate the Laplace transform of the impulse response as well as the actual impulse response

## Matlab

Implement the systems on slides 10 & 12 in Simulink and verify their responses by exact calculation.

Note that `roots()` is a Matlab function that will calculate the roots of a polynomial expression